

9. Explain the following :

(a) Instruction Format

(b) I/O Channel

8

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Roll No.

97666

**BCA 2nd Semester
Examination – April, 2018**

LOGICAL ORGANIZATION OF COMPUTER

Paper : BCA-107

Time : Three Hours] [Maximum Marks : 80

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Question No. 1 is compulsory. Attempt four questions by selecting one question from each Unit.
All questions carry equal marks.

1. (a) What is Interrupt structure ? $2 \times 8 = 16$
(b) What is an Instruction Set ?
(c) What are applications of ROM ?
(d) What is an Instruction Cycle ?
(e) What is an I/O Interface ?
(f) What characterizes optical storage devices ?

- (g) What is an Instruction Format ? State their significance. 8
- (h) What is D flip-flop ? 8

UNIT - I

2. (a) What are Excitation Tables ? How are these relevant ? Draw Excitation Table for RS and JK flip-flop. 8
- (b) What are race conditions ? How does Master-Slave flip-flop overcome race conditions ? Illustrate its working. 8

3. Explain the following :

- (a) Clocked RS flip-flop 8
- (b) State Diagram 8

UNIT - II

4. (a) What is Modulo-10 counter ? How do you design it ? Explain its working through block diagram. 8
- (b) Design a 4-bit shift register and outline the procedure for serial to parallel conversion and vice-versa. 8

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5. Explain the following :

- (a) Up-Down Counter 8
- (b) Asynchronous Sequential Circuit 8

UNIT - III

6. (a) What are I/O device controllers ? How these work ? Illustrate their working. 8
- (b) What is Internal memory ? How is it different from External memory ? What are important memory parameters ? Illustrate. 8

7. Explain the following :

- (a) Flash Memory 8
- (b) Semiconductor RAM 8

UNIT - IV

8. (a) What are Addressing Modes ? What are different Addressing Modes ? Illustrate through examples. 8
- (b) What is DMA technique ? How is it different from Interrupt-controlled I/O technique ? Illustrate their working. 8

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